MIL-C-37040A
28 September 1978
SUPERSEDING
MIL-C-37040 (DSA-DM)
12 March 1974

MILITARY SPECIFICATION

CENTRIFUGE, LABORATORY, SMALL, 115 Volt, 60 Hz, AC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of small table model centrifuge suitable for use in military medical laboratories having limited space, where the character of work does not require centrifuging large volumes.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-M-7298	- Manual, Technical: Commercial Equipment.
MIL-P-116	- Preservation-Packaging, Methods Of.
MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, CODE ATT, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6640

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STANDARDS

FEDERAL

FED-STD-66 - Steel: Chemical Composition and Hardenability.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C73.11 - Dimensions of Plugs and Receptacles - General Purpose, 125 Volts, 15 Amperes, 2-Pole, 3-Wire, Grounding Type.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018).

UNDERWRITERS' LABORATORIES, INC. (UL)

UL-73 - Motor-Operated Appliances.

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingsten Road, Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, L.I., NY 11746; 1655 Scott Boulevard, Santa Clara, CA 05050).

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 The contractor shall submit proof, prior to delivery, that the items to be delivered under the contract conform to the applicable requirements of UL 73. Proof shall consist of one of the following:

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a. UL label or UL listing of the product, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the UL report directly to the contracting officer, upon the latter's request.

b. Laboratory report from a qualified independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The qualification of such laboratory shall be determined by the Government, which reserves the right to require a facilities report and to conduct a facilities survey. Any laboratory report submitted shall include actual test methods, data taken, and results of tests. Detailed findings of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample. This section shall be supplemented with construction information such as materials used, spacings between electrical components, types of bushings, photographs, wiring diagrams, sketches, and parts list, all appropriately identified.

3.1.2 Compliance with above paragraph does not excuse compliance with all other requirements of the contract and specifications.

3.2 Material. Material shall be as specified herein. When material is not specified, it shall be of a quality which meets the functional requirements of the centrifuge.

3.3 Construction. The centrifuge shall consist of a guard bowl, a cover, a motor, a base and a speed control device.

3.3.1 Guard bowl. For purposes of this specification, the guard bowl is the chamber which houses the drive spindle, head, trunnions and shields. The walls of the guard bowl shall be fabricated of steel which is at least 0.048 inch (No. 18 U.S. Standard Gauge) thick, and commercial tolerances shall apply. Top view of the guard bowl shall be circular. Inner circumferences of the bowl shall be such that they provide a chamber which is large enough to insert the head, trunnions and other accessories listed in 3.6.

3.3.2 Cover. Cover shall be located above the guard bowl and shall be fabricated of steel which is at least 0.048 inch (No. 18 U.S. Standard Gauge) thick, and commercial tolerances shall apply. Cover shall be secured to the chamber by means of a hinge. Center of cover shall have a hole which provides for safe use of speed counter.

3.3.3 Motor. Motor shall be the universal speed type which operates on 115 volt, 50/60 Hertz, alternating current.

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3.3.4 Drive spindle. The drive spindle shall be vertically mounted and shall be an integral extension of the motor shaft. Drive spindle shall be adequate size to support static dynamic forces encountered in the operation of the centrifuge.

3.3.5 Base. Base shall be fabricated of cast iron and shall be of such design and size that it supports the guard bowl and houses the motor. Bottom perimeter shall have at least three equally spaced rubber feet which provide cushioning and stability.

3.3.6 Speed control device. Speed control device shall be an autotransformer or a rheostat and shall provide for not less than seven steps of speed control and an "off" position. The device shall be legibly identified and located so as to be convenient to the operator.

3.4 Dimensions. Overall height with cover closed shall be not more than 13 inches. Outer diameter of guard bowl shall be not more than 14 inches.

3.5 Cord and plug. The centrifuge shall have a three-wire electrical cord equipped with a grounding plug in accordance with ANSI C73.11. Cord shall be a flexible, heavy duty type and shall be at least five feet long when measured from the outside of the centrifuge.

3.6 Accessories. Centrifuge shall be supplied with the following accessories, which shall be interchangeable with all centrifuges procured under a specific contract.

Quantity	Item
1	6-place horizontal head with trunnion rings for 15-ml. shields.
6	15-ml. corrosion-resisting steel shields with rubber cushions. Steel shall be in accordance with FED-STD-66, AISI 300 series.

The heads shall be dynamically balanced. The shields shall be weighed and matched to 0.5 gram; weight in grams shall be stamped on each shield.

3.7 Finish. Centrifuge shall be finished with a durable baked-on enamel.

3.8 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the

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Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the contracting officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

3.9 Performance.

3.9.1 Relative centrifugal force. Centrifuge shall attain a force between 1470 and 2000 gravities when tested as specified (see 4.4.1).

* 3.9.2 Operation. When tested as specified, the centrifuge shall operate without any type of failure or tendency to creep. There shall be no evidence of excessive vibration or excessive noise that would indicate misalignment or imbalance of the motor or head (see 4.4.2).

* 3.10 Nameplate. Nameplate shall be permanently and legibly marked with the National Stock Number, item identification, contract or purchase order number, model or catalog number, serial number, manufacturer's name or trademark, voltage, type of current, watts input or amperage. Nameplate shall be secured to the centrifuge by means of screws or rivets.

* 3.11 Workmanship. Workmanship shall be first class throughout. Centrifuge and accessories shall be free of defects indicated in table III or other defects which detract from their apperance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I.

TABLE I. Sampling for examination.

	Inspection level	AQL (percent defective)
For visual examination Major defects	II	1.0
For dimensional examination	S-1	2.5

* 4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II. Acceptance number shall be zero.

TABLE II. Sampling for tests.

Characteristic	Requirement	Test procedure	Inspection level
Relative centrifugal force	3.9.1	4.4.1	S-1
Operation	3.9.2	4.4.2	100%

4.3 Examination. Centrifuge shall be examined to determine compliance with all requirements in this specification. Examination shall include, but shall not be limited to, the defects listed in table III.

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TABLE III. Classification of defects.

Categories	Defects
Major	
101	Component or accessory missing.
102	Component not secured.
103	Centrifuge or accessories not free of sharp edges or burrs.
104	Finish not free of chipping, cracking or peeling.
105	Centrifuge, components or accessories not free of rust spots.
106	Service data missing.
107	Numerals and other markings on components and accessories not complete, correct, legible or permanent.
108	Identification marking not complete, correct, legible or permanent.

* 4.3.1 Dimensional and weight examination. The centrifuge and accessories shall be examined for defects in dimensions and weight. Any dimension or weight not within the tolerance specified herein shall be considered as a defect.

* 4.3.2 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Relative centrifugal force. Head shall be placed on drive shaft and shall be filled with 15-ml. centrifuge tubes, which are filled with water and are in shields. Centrifuge shall be activated, and test procedures shall be in accordance with service data (see 3.8). Speed shall be set at maximum. Rotating speed shall be measured by a stroboscope or tachometer of known accuracy, and relative centrifugal force at the tip of the 15-ml. centrifuge tube shall be calculated in accordance with procedures in service data and in accordance with the following formula:

$$RCF = 0.00001118 \times r \times N^2$$

RCF = relative centrifugal force in gravities.
r = rotating radius to tip of centrifuge tube in centimeters.
N = rotating speed in revolutions per minute.

Test shall be run at 115 volts, 60 Hz.

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4.4.2 Operation. With a head placed on the drive shaft, an operational check shall be performed on the centrifuge at maximum speed to demonstrate that the centrifuge is capable of proper operation.

* 5. PACKAGING

5.1 Preservation-packing. Preservation-packing shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning. Each centrifuge along with accessories shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Each centrifuge along with accessories shall be dried as specified in MIL-P-116.

5.1.1.3 Preservative application. A preservative shall not be used.

5.1.1.4 Unit packaging. Each centrifuge along with accessories shall be packaged in accordance with MIL-P-116, method III, within a box conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.1.2 Level C. Each centrifuge along with accessories shall be packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Four unit packages shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-585, class 3, style 3; PPP-B-601, overseas type; or PPP-B-621, class 2. The box shall be lined with a case liner conforming to MIL-L-10547. Closure and sealing of the liner shall be as specified in the appendix of the liner specification. Closure, and strapping of the box shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B. Four unit packages shall be packed in a shipping container conforming to PPP-B-585, class 1, style 3; PPP-B-601, domestic type or PPP-B-621, class 1. Closure shall be as specified in the appendix of the applicable box specification.

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5.2.3 Level C. The packaged centrifuges shall be packed in shipping containers in a manner that will afford adequate protection against damage during shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carriers' rules and regulations.

5.3 Marking. Each unit package, intermediate package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following.

(a) Title, number, and date of this specification.
(b) National Stock Number (NSN).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following item shown in the Federal Supply Catalog.

National Stock Number	Item Identification
6640-00-930-9034	CENTRIFUGE, LABORATORY, SMALL, 115 Volt, 60 Hz, AC

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
Defense Logistics Agency - DM

Project Number 6640-1055

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-37040A CENTRIFUGE, LABORATORY, Small, 115 Volt, 60 Hz, AC

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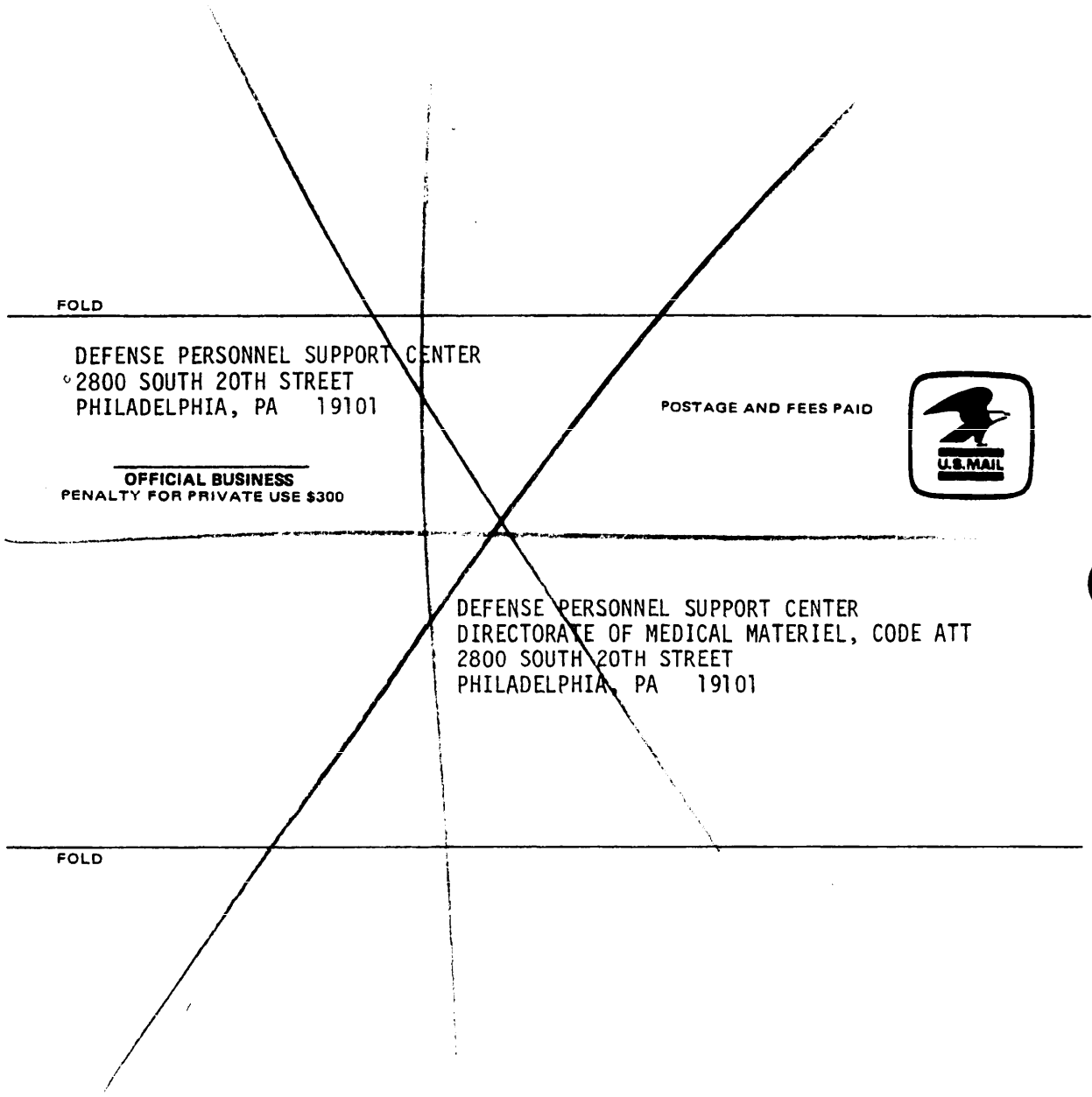
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